

Article

Process Concept of a Waste-Fired Zero-Emission Integrated Gasification Static Cycle Power Plant

Augusto Montisci ^{1,*}  and Aiman Rashid ^{1,2} 

¹ Department of Electrical and Electronic Engineering, University of Cagliari, 09123 Cagliari, Italy; aiman.rashid@unica.it

² Department of Industrial Engineering, University of Florence, 50139 Florence, Italy

* Correspondence: augusto.montisci@unica.it; Tel.: +39-320-437-3053

Abstract

The layout of an urban waste-fired zero-emission power plant is described in this paper. The principle layout, which is based on similar coal-fired plants retrieved from the literature, integrates gasification with a power-generation section and implements two parallel conversion processes: one relies on the heat developed in the gasifier and consists of a thermoacoustic-magnetohydrodynamic (TA-MHD) generator; the other involves treating syngas to obtain almost pure hydrogen, which is then fed to fuel cells. The CO₂ derived from the oxidation of Carbon is stocked in liquid form. The novelty of the proposed layout lies in the fact that the entire conversion is performed using static equipment. The resulting plant prevents the release of any type of emissions in the atmosphere and increases mechanical efficiency, compared to traditional plants—thanks to the absence of moving parts—resolving, nonetheless, the ever-increasing waste-related pollution issue. A case study of a Union of Municipalities in Southern Lebanon is considered. The ideal cycle handles 65 tons/day of urban waste and is capable of generating 7.71 MW of electric power, with a global efficiency of 52.39%.

Keywords: magnetohydrodynamics; thermos-acoustics; fuel cells; Waste-to-Energy plant; zero-emission power plant



Academic Editor: Maurizio Volpe

Received: 21 May 2025

Revised: 18 June 2025

Accepted: 20 June 2025

Published: 24 June 2025

Citation: Montisci, A.; Rashid, A. Process Concept of a Waste-Fired Zero-Emission Integrated Gasification Static Cycle Power Plant. *Sustainability* **2025**, *17*, 5816. <https://doi.org/10.3390/su17135816>

Copyright: © 2025 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

1. Introduction

By the year 2050, it is projected that a significant majority of the global population, estimated at 66%, will be concentrated in urban areas, marking a substantial rise from the current approximate proportion of 54% [1]. This demographic shift indicates the potential addition of approximately 2.4 billion individuals to the global urban populace. Consequently, such urbanization trends will inevitably instigate a substantial expansion of existing urban landscapes and necessitate the establishment of novel urban environments. Notably, despite occupying less than 2% of the Earth's surface, cities disproportionately consume more than 75% of the world's available natural resources. The United Nations Environment Programme (UNEP) anticipates that the material consumption attributed to urban areas will soar to approximately 90 billion tons by 2050, a notable escalation from the 40 billion tons recorded in 2010. These resources encompass critical components such as primary energy, raw materials, fossil fuels, water, and food [2].

As a consequence, urban areas are poised to confront formidable challenges concerning growth, performance, competitiveness, and the overall quality of residents' lives [1]. The deterioration of living standards arising from issues such as waste management, resource

scarcity, air pollution, and traffic congestion, not only give rise to concerns about human health but also contribute to the degradation of aging public infrastructure, exemplifying the complexity generated by rapid urbanization [3]. To effectively tackle these multifaceted problems, the concept of smart cities has emerged as a promising avenue for exploration and potential resolution.

The central emphasis of this paper revolves around the reduction of pollution deriving from waste, specifically targeting waste materials that are not amenable to recycling or deemed economically unfeasible for recycling purposes. The novelty of this study is the concept of a Waste-to-Energy (WtE) plant, completely free of atmospheric emissions, thus significantly reducing the external costs of the plant. In addition, all devices used in the different processes operate without moving solid parts. This constitutes a great advantage in terms of loss reduction, maintenance costs, efficiency of thermodynamic cycles, availability, and plant lifespan. In addition, the technologies used are characterized by a high degree of scalability, as well as high power density, so they have the additional advantage of requiring less land consumption than conventional technologies.

The present work is inspired by the technology for zero-emission coal-fired power plants, which integrate coal gasification with a combined gas–steam cycle. In fact, while the main objective in waste management is to prevent waste production, in coal-fired power plants, the problem of emissions cannot be avoided. In the case study outlined in this work, a significant proportion of this waste cannot be recycled, especially in view of remediation works of numerous existing landfills, where waste presents the risk of several mutual contaminations among different fractions. The most natural solution is thermal treatment, and therefore the same problem of emissions arises as in coal-fired power plants. Compared to these options, the proposed solution has some differences, mainly due to the much lower power level. For this reason, while the same process is adopted for fuel gasification and syngas treatment, the power section is completely different, as the combined cycle is replaced by a fuel cell.

The paper is organized into four sections. In Section 2, the technological background is given. In Section 3, the technologies implemented in the power plant are described separately in detail. In Section 4, the layout of the proposed WtE plant is described. Section 5 provides numerical results, while in Section 6, a preliminary economic analysis is presented. In Section 7, some comments and conclusions are given.

2. Technological Background

2.1. Waste Incineration Treatments

Waste incineration has long been a cornerstone of modern waste management strategies, offering both volume reduction and energy recovery. As global waste generation continues to rise, the choice of incineration technology plays a critical role in balancing efficiency, environmental impact, and economic feasibility. This section examines the most prominent waste incineration methods, evaluating their strengths and limitations based on current research.

2.1.1. Mass Burn Incineration: The Conventional Approach

Mass burn incineration remains the most widely adopted method for treating municipal solid waste (MSW). In this process, unprocessed waste is combusted in large furnaces, generating heat that can be converted into electricity or district heating. One of its key advantages is its ability to reduce waste volume by up to 90%, significantly alleviating landfill pressures [4]. Additionally, mass burn plants contribute to energy security by supplying power grids, particularly in urban centres where waste availability is high [5].

However, this method is not without its drawbacks. The combustion of mixed waste can release harmful pollutants, including dioxins, furans, and particulate matter, necessitating advanced flue gas cleaning systems [6]. Critics also argue that mass incineration may compete with recycling efforts, as it requires a steady waste stream to remain economically viable, potentially discouraging waste minimization policies.

2.1.2. Fluidized Bed Incineration: Enhanced Combustion Efficiency

For waste streams with high moisture or variability, Fluidized Bed Incineration (FBI) offers a more efficient alternative. By suspending waste particles in a heated bed of sand or similar material, FBI ensures uniform combustion at lower temperatures (800–900 °C), reducing nitrogen oxide (NO_x) emissions compared to mass burn systems [7]. This technology is particularly effective for treating sewage sludge and industrial waste, where controlled combustion is crucial [5].

Despite its advantages, FBI requires extensive waste pre-processing, such as shredding, to ensure consistent feedstock size. The abrasive nature of the bed material also leads to higher maintenance costs, and operational stability can be compromised by fluctuations in waste composition [8].

2.1.3. Modular Incineration: Small-Scale Solutions

In remote or underserved regions, modular incinerators provide a decentralized waste treatment option. These compact units are particularly valuable for medical waste disposal, where immediate treatment is necessary to prevent contamination [9]. Their lower infrastructure requirements make them suitable for areas lacking large-scale waste management facilities.

Yet, modular systems suffer from lower energy recovery rates and higher per-ton operating costs compared to centralized plants [10]. There is also a risk of inadequate emission controls in regions with lax environmental regulations, raising concerns over air quality impacts.

2.1.4. Advanced Thermal Treatments: Gasification and Plasma Arc

At the cutting edge of WtE technology, gasification and plasma arc incineration convert waste into syngas (a mixture of carbon monoxide and hydrogen) or inert slag through extremely high temperatures (1000–12,000 °C). These methods boast near-complete waste destruction, with residual slag often repurposed in construction materials [9]. The oxygen-starved environment in gasification also minimizes dioxin formation, offering a cleaner alternative to conventional incineration.

However, the high capital and operational costs of these systems have hindered widespread adoption. Their technical complexity demands skilled personnel, and scalability remains a challenge for municipal applications [11].

2.1.5. Co-Incineration: Synergy with Industrial Processes

An alternative approach involves co-incinerating waste-derived fuels (such as refuse-derived fuel, RDF) in cement kilns or power plants. This method not only reduces fossil fuel consumption—with substitution rates reaching up to 30% in some cases—but also leverages existing industrial infrastructure [12].

The primary concern with co-incineration is the potential release of heavy metals into cement products, which could pose long-term environmental and health risks [13]. Strict feedstock quality control is therefore essential to mitigate contamination.

In countries where landfills are an environmental and public health issue, municipal waste is often incinerated to reduce volume, and this is increasingly done alongside energy recovery, generating electricity and heat (e.g., for regional domestic hot water systems) [14].

These activities, in turn, cause major problems in the environment due to emissions of all kinds of pollution and their impact on ecosystems. Problems of emissions and greenhouse gases and 135 pollutants related to waste include landfills etc. Each form of waste incineration technology presents a unique set of trade-offs between efficiency, environmental impact, and cost. While mass burn incineration dominates the market, emerging technologies like gasification and fluidized beds offer promising avenues for cleaner energy recovery. Finally, the thermal treatment of waste can be employed, but the issues caused by harmful gases and solid ash must be handled properly. From this point of view, some techniques, developed over the past century to prevent emissions resulting from the use of coal [15], can be applied in this context.

2.2. *Advancements in the Coal Handling*

The coal handling industry, long considered a traditional sector, is undergoing a quiet revolution. As global energy systems transition toward cleaner alternatives, coal remains a critical energy source in many regions, necessitating innovations that improve efficiency, reduce environmental impact, and enhance worker safety. Recent technological advancements are reshaping coal handling—from mining and transportation to processing and utilization—offering both promising solutions and new challenges.

2.2.1. The Rise of Smart Coal Handling

One of the most transformative developments is the integration of artificial intelligence (AI) and automation into coal sorting and processing. Traditional methods relied heavily on manual labor for quality control, but AI-driven optical sorting systems now use machine learning algorithms to classify coal based on purity, calorific value, and ash content. These systems, equipped with high-resolution cameras and robotic arms, can separate coal from waste rock with remarkable precision [16]. The benefits are clear: higher-quality coal output, reduced human exposure to hazardous dust, and improved compliance with emissions regulations. However, the high initial investment and ongoing maintenance of these AI systems pose financial barriers, particularly for smaller mining operations [17].

2.2.2. Protecting Workers and the Environment

Coal dust has long been a pervasive issue, contributing to respiratory diseases among workers and increasing the risk of explosions in handling facilities. Modern dust suppression technologies, such as advanced fogging systems and electrostatic precipitators, have significantly mitigated these risks. Chemical binders, for instance, can now stabilize coal piles, reducing airborne particulate matter by up to 90% [18]. These innovations not only enhance workplace safety but also help companies meet stringent environmental regulations. Yet, challenges remain. In water-scarce regions, fogging systems put strain on local resources, while chemical treatments introduce additional operational costs [19].

2.2.3. IoT and Energy-Efficient Conveyors

The transportation of coal within mining and processing facilities has also seen notable improvements. Traditional conveyor belts, once a major source of energy loss and mechanical failure, are now being replaced with IoT-enabled systems. These “smart conveyors” use embedded sensors to monitor belt tension, alignment, and wear in real time, enabling predictive maintenance and reducing unplanned downtime [20]. Some systems even incorporate regenerative drives that recover and reuse energy, cutting electricity consumption by nearly a third [21]. While these advancements boost efficiency, they also introduce cybersecurity vulnerabilities, as interconnected systems become potential targets for hacking [22].

2.2.4. Advanced Coal Washing and Gasification

Beyond handling, innovations in coal processing aim to reduce its environmental footprint. High-efficiency cyclones and froth flotation techniques now remove a greater share of impurities, yielding cleaner coal with higher energy content [23]. This not only improves combustion efficiency but also reduces slag buildup in power plant boilers, extending their operational life. Meanwhile, coal gasification and coal-to-liquids (CTL) technologies are gaining traction as methods to produce synthetic fuels with lower emissions. When paired with carbon capture and storage (CCS), these processes offer a pathway to decarbonize coal use [23]. However, the high capital costs and substantial water requirements of gasification plants limit their widespread adoption, particularly in developing economies [24].

2.2.5. Integration and Sustainability

The coal handling industry stands at a crossroads. While technological advancements have undeniably improved efficiency and safety, they also highlight the sector's ongoing reliance on finite resources and its environmental trade-offs. Future progress may hinge on integrating renewable energy sources into coal handling operations—such as using solar power to run conveyor systems—or further refining carbon capture technologies to mitigate emissions.

The industry's evolution reflects broader tensions between immediate operational needs and long-term sustainability goals. As innovations continue to emerge, their success will depend not only on technical feasibility but also on economic viability and regulatory support. The challenge for stakeholders is to adopt these advancements in a way that balances productivity with environmental and social responsibility.

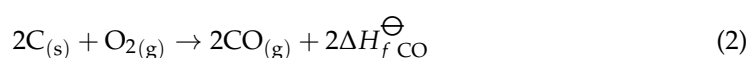
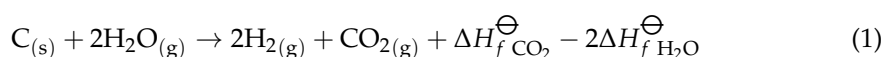
In this paper, a Zero-Emissions Integrated Gasification Static Conversion (ZE-IGSC) waste treatment plant is proposed, which integrates gasification with a power generation section. The entire plant is devoid of moving parts and the emission of any gases in the atmosphere is avoided. To this end, energy conversion is performed by a series of static devices, such as the gasifier, thermoacoustic resonators [25], magnetohydrodynamic (MHD) electrical generators, and fuel cells. The resultant CO₂ is stored in liquid state as a marketable byproduct of the whole process [26,27].

3. Materials and Methods

3.1. Gasification

The idea of integrating gasification with a thermoelectric conversion section has been proposed in the past for coal-fired power plants in order to enhance global efficiency and mitigate emissions [28]. There are basically three technologies of gasifiers: Moving Bed, Fluidized Bed, and Entrained Flow [29]. The third is characterized by the highest efficiency, but its operation is difficult. The charge can consist of dry fuel or slurry, where the fuel is premixed with water. In both cases a certain percentage of water must be injected into the gasifier together with the fuel and the oxygen, so that two kinds of reaction take place, involving water and oxygen, respectively.

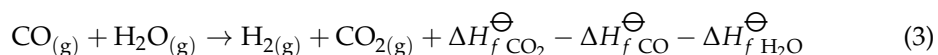
Here the equilibrium equations of the two reactions are reported [30–32]:



3.2. Water-Gas Shift Reactions

The water–gas shift reaction (WGSR) is performed downstream of gasification to change the $H_2:CO$ ratio. The WGSR makes use of equal moles of carbon monoxide (CO) and steam ($H_2O_{(g)}$) in the mixture, and it is weakly exothermic [33].

In the WGSR process, the water molecule is separated as follows [34]:



Because WGSR is moderately exothermic [30–32], the equilibrium constant of the reaction decreases as temperature rises, but high yield is achieved with low temperatures. If the quantity of steam is increased with respect to the stoichiometric equilibrium, the conversion is advanced. Nonetheless, at low temperatures, the stoichiometric and dynamic equilibrium helps in the growth of products with the domination of reaction kinetics [33,35].

The WGSR works perfectly between 477 K and 755 K in the presence of several catalysts. The total number of moles of reactants is the same as products, resulting in a very minimal effect of pressure on the reaction. Typically, a very large amount of moisture is available in the scrubber syngas, due to slurry-fed gasifier, which is adequate to push the WGSR to attain the essential ratio of $H_2:CO$. However, in some such cases, some proportion of the syngas feed is detoured, to prevent product excess. If the gasifier is dry-fed, supplementary steam ($H_2O_{(g)}$) is added before the WGSR begins for appropriate syngas output [35].

3.3. Integrated Gasification Combined Cycle

Integrated Gasification Combined Cycle (IGCC) technology adds fewer emissions to the atmosphere and offers more flexibility in terms of fuel selection compared to traditional coal-based power generation technologies [15]. IGCC is a next-generation thermal power technology, which partially oxidizes solid feedstock (biomass, coal, lignite, etc.) with oxygen (O_2) and steam ($H_2O_{(g)}$) to produce syngas, or synthesis gas [36–38]. Fuel gas, or synthesis gas, is a mixture of mainly three gasses—hydrogen (H_2), carbon monoxide (CO), and some carbon dioxide (CO_2)—which could be used as an intermediate in the conversion of biomass into fuel [39]. Conventionally, when power is generated without the collection of carbon with IGCC design, the synthesis gas is cleansed of hydrogen sulfide (H_2S) and dust, and then fed to a Gas-Steam Combined Cycle system for power production. When they are present, the technologies for carbon capture and storage (CCS) are used to take advantage of the high concentration of the CO_2 [38]. IGCC has the advantage not only of capturing CO_2 but also of not incurring many efficiency penalties for the power plant, in terms of CAPEX and OPEX [36]. One benefit of using IGCC is that it has the ability to effectively capture carbon dioxide before combustion, with benefit of enhanced removal efficiency compared to typical power plants fired with pulverized coal [15]. The estimated increase in efficiency attributed to the deployment of large-type IGCC systems is about 15%, in addition to reductions in carbon dioxide emissions compared to typical coal-fired thermal power systems [28,37].

3.4. Waste to Energy

Most developed, as well as developing, countries are irrefutably facing the dire problem of handling, treating, and compacting waste due to mismanagement, which results in general dangers and threats to biodiversity and ecosystems, and specific threats to the atmosphere and environment. Well-managed municipal solid waste (MSW) is among the sources of waste to energy (WTE). It has the capability to produce biogas that can combinedly generate heat and power through the application of WTE processes. Such

techniques must be conceptualized based on composition and assessment, keeping the financial value of the waste as a criterion. However, selecting the appropriate technology for WTE is a difficult task as the generation of waste is constantly affected by the producer's season, region, and socioeconomic level [40].

3.5. Magnetohydrodynamic Electric Generators

The conversion efficiency for the performance of the steam plants has already achieved its upper limit of 40%, and the only workable solution to enhance it further is to combine steam cycles with various conversion systems. The magnetohydrodynamic (MHD) generator is intrinsically suitable for this application, because of its operation at high temperatures [41]. The MHD generator is a static power conversion technology which generates electricity by exploiting a conducting fluid (plasma), which flows through a magnetic field. When the plasma moves in the presence of the magnetic field, it behaves as a mobile electrical conductor, and it becomes seat of an electromotive force [42].

The generic charge q in the plasma flowing in the induction field $\mathbf{B}$ is subject to the Lorentz force $\mathbf{F}$ (Figure 1) according to the equation:

$$\mathbf{F} = q(\mathbf{v} \times \mathbf{B}) \quad (4)$$

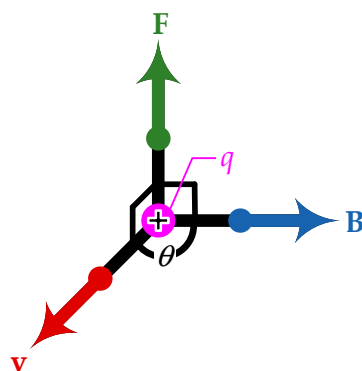


Figure 1. Charge q moving at velocity $\mathbf{v}$ in magnetic field $\mathbf{B}$, experiences a Lorentz force $\mathbf{F}$ normal to the plane containing $\mathbf{v}$ and $\mathbf{B}$.

Two electrodes at direct contact with the plasma, arranged perpendicularly with respect to both the flow and the induction field and positioned on opposite walls of the duct, intercept the charges providing an interface with the external circuit, obtaining a source of direct current, with density $\mathbf{J}$ (Equation (5)).

$$\mathbf{J} = \sigma(\mathbf{v} \times \mathbf{B}) \quad (5)$$

where σ is the electric conductivity of the plasma. The yielded electric energy, net of losses, is equal to the reduction of enthalpy of the fluid stream.

The equations for fluid mechanics used in MHD are the mass and the momentum conservation equations (see Equations (6) and (7)) with the addition of the electromagnetic force, depending on the current density and on the magnetic field according to Equation (4):

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0 \quad (6)$$

$$\frac{\partial}{\partial t}(\rho \mathbf{v}) = \mathbf{v} \cdot \nabla(\rho \mathbf{v}) = -\nabla p + \nabla \cdot \boldsymbol{\tau} + \mathbf{J} \times \mathbf{B} + \mathbf{F}_v \quad (7)$$

where ρ is the fluid mass density, p is the pressure, $\boldsymbol{\tau}$ is the stress tensor, and $(\rho \mathbf{g})$ is the specific gravity force, if $\mathbf{g}$ is gravitational acceleration. The gravitational effects are

neglected because of the height differences between any pair of points compared to the weight of the gas. The stress tensor could be expressed by the linear constitutive equation.

$$\boldsymbol{\tau} = \zeta(\nabla \cdot \mathbf{v})\mathbf{I}_d + \eta \left(\nabla \mathbf{v} + (\nabla \mathbf{v})^T - \frac{2}{3}(\nabla \cdot \mathbf{v})\mathbf{I}_d \right) \quad (8)$$

where $\mathbf{I}_d$ is the identity tensor, ζ is the bulk viscosity, and η is the fluid dynamic viscosity. The first term in the right-hand side of Equation (8) is usually neglected for incompressible fluids or monoatomic gases like noble gases at standard temperature and pressure or, in general, like all the gases at a sufficiently high temperature. Therefore, the most general form of the momentum Equation for compressible fluids becomes:

$$\frac{\partial}{\partial t}(\rho \mathbf{v}) + \mathbf{v} \cdot \nabla(\rho \mathbf{v}) = -\nabla p + \eta \left(\Delta \mathbf{u} + \frac{1}{3} \nabla(\nabla \cdot \mathbf{u}) \right) + \mathbf{J} \times \mathbf{B} \quad (9)$$

The plasma thermodynamic behaviour could be described by the ideal gas state Equation (10) since the proposed study deals with noble gases or air at standard temperature and pressure conditions.

$$p = \rho \mathcal{R} T \quad (10)$$

where $\mathcal{R}$ is the specific gas constant and T is the temperature of the gas.

The energy transfer could be modelled through a global energy balance:

$$\rho \frac{d(e)}{dt} = -\nabla \cdot (p \mathbf{u}) + \nabla \cdot (k \nabla T) + \mathbf{u} \cdot (\nabla \cdot \boldsymbol{\tau}) + \mathbf{J} \cdot \mathbf{E} + S_{\text{int}} + Q_{\text{wall}} \quad (11)$$

where e is the total energy per volume unit (sum of kinetic and internal specific energy), k is the thermal conductivity, S_{int} a generic internal power source, and Q_{wall} is the wall heat transfer per volume unit and time unit. Equation (11) will be simplified considering the adiabatic flow assumption:

$$\frac{d}{dt} \left(\frac{p}{\rho^\gamma} \right) = 0 \quad (12)$$

where $\gamma = c_p/c_v$ is the heat capacity ratio between specific heat at constant pressure and specific heat at constant volume. Equation (12) is the energy Equation written for the simplest possible functioning conditions and could be used replacing the overall energy balance.

The power of an MHD generator for each cubic meter of the channel volume is given by Equation (13) [27]:

$$P_{\text{el}} = \mathbf{J} \cdot \mathbf{V}_0 = \sigma v^2 B^2 K(K-1) \quad (13)$$

where $K = V_0/E_{\text{ind}} = V_0/(\mathbf{v} \cdot \mathbf{B})$ is the load factor (ratio between open-circuit voltage and induced voltage). It can be noted that, since the current density $\mathbf{J}$ flowing through the load and the open circuit voltage are aligned, the dot product between $\mathbf{J}$ and $\mathbf{V}_0$ is electrical power P_{el} between modules (Equation (13)). Therefore, the power P_{el} is proportional to the electrical conductivity σ , the square of the gas velocity $\mathbf{v}$, and the square of the intensity of the magnetic field $\mathbf{B}$. In order to reach a competitive operability with existing electric generators, the MHD system needs to have a high conductive gas, at high temperature, high speed, compatibly with the increase of the turbulence, and an extremely intense magnetic field (5–7 tesla), coherently with the costs for the necessary superconducting electromagnets [43–45].

Typically, an electrically conductive high-pressure gas is produced by combustion of fossil fuels. Unfortunately, the most common gases do not ionize significantly at temperatures obtainable through chemical reactions with fossil fuels. This requires doping the hot gases with small amounts of easily ionizable materials, such as alkali metals. Materials

such as potassium and cesium have very low ionization potentials that tend to ionize at temperatures achievable by combustion in air. Recovery and reuse of doping materials from the MHD channel exhaust system are fundamental both in terms of economic issues and pollution prevention.

The application of MHD generators is especially suitable for large-scale power production, as these work as volumetric expansion engines. Since the MHD generators only consist of ducted components, and no high-stressed moving parts are present, their reliability as a power generation system is pronounced at high temperatures [41].

Types of MHD Generators

There are three fundamental methods to MHD power generation: open cycle, closed cycle plasma and closed liquid-metal systems. In open cycle MHD systems, the working fluid is emitted in the atmosphere after electricity generation [46–50], the closed cycle MHD system features an extra step in that it recycles the working fluid in order to make use of the thermal energy present in the fluid as it exits the MHD chamber (after the electricity has been generated by the MHD system) [51], whereas in closed liquid-metal MHD systems, the liquid metals are used as the electrically conducting fluids; that is why they are called liquid-metal MHD generators [52,53]. Liquid metals can be used as conducting fluids in MHD generators, because the metals have extremely high electrical conductivity. Another benefit of generating electricity with liquid-metal MHD generators is they can work at lower temperatures compared to other MHD generators, which require high temperatures for the production of plasma. The normal working principle is that the liquid metal is accelerated with a thermodynamic pump or fused with the driving gas and then detached from the driving gas before the liquid metal passes through the MHD channel [42,52].

MHD generators offer several advantages in terms of efficiency and simplicity. Since they directly convert the energy of the fluid into electricity, without intermediate mechanical conversion, they can achieve higher overall efficiency compared to traditional power generation methods. Additionally, MHD generators can operate at high temperatures, increasing the thermodynamic limit of efficiency [42].

In the generators that use liquid metals, the challenges are of a different nature. Indeed, conductivity is high at any temperature, rendering seeding unnecessary, and lower magnetic fields are suitable, meaning superconducting coils are not necessary. On the other hand, the enthalpy of the combustion gas must be transferred to the liquid metal, and such enthalpy cannot be in the form of heat, because the density variation in temperature is negligible for liquids. For this reason, in cases where the primary source is heat, an intermediate transformation is needed before the use of a liquid-metal MHD generator.

3.6. Thermoacoustic Converter

A Thermo-Acoustic Resonator (TAR) is a device which statically converts heat into vibration power, leveraging on a gradient of temperature. In Table 1 three examples of TAR technologies are reported. The main principle of a thermoacoustic heat engine is to convert thermal power into acoustic power without any mechanical moving parts, while acoustic power is converted to electricity by linear alternators. Inexpensive electricity in developing countries can be provided by using simple fabrications of such thermoacoustic electricity generators (TAG). The waste heat present in flue gases can be recovered by thermoacoustic energy conversion processes to generate electricity as by-product [54]. When a temperature gradient is imposed in the direction of the acoustic wave propagation, interactions between the gas undergoing acoustic excitation and a porous medium give rise to thermoacoustic effects [25].

Table 1. Technologies of thermoacoustic and magnetohydrodynamic power generation/Overview of renewable energy generation technologies.

Technology/Process	Power	Advantages	Disadvantages	Refs
Thermoacoustic generator				
Standing wave (wf: air)	0.21 W 0.084%	Novel cold heat exchanger design for thermoacoustic engine (TAE) is introduced.	Helium gas can be used to improve efficiency.	[55]
Travelling wave (wf: He-Ar mixture)	10.6 W 1.51%	Three-stage looped thermoacoustic electric generator	The thermal-to-electric efficiency stabilizes at around 1.5% when hot temperature is in the range of 120–170 °C.	[56]
Travelling wave (wf: helium)	3.46 kW 18.4%	Three-stage traveling-wave thermoacoustic electric generator (heat engine and linear alternators)	The acoustic impedance of the linear alternator can be greatly affected by the electric capacitance and resistance, thereby the system performance can be greatly changed.	[57]

wf: working fluid.

The leading sustainable sources of energy include low-grade thermal energy, such as solar thermal energy, geothermal energy, ocean thermal energy, and waste heat. Thanks to exceptional characteristics of extreme reliability and environment-friendliness, thermoacoustic generators are among the most appealing and promising energy conversion solutions to recover low-grade thermal energy [56]. Additionally, helium and nitrogen are the working substances used in TAG, which are environmentally friendly gases [57].

The thermoacoustic resonator typically consists of a resonant circuit, a stack of porous materials, a high-temperature heat source, and a low-temperature heat sink [25,55]. Vibration is triggered by the thermal gradient along the wall of the resonator. The efficiency of the thermomechanical conversion is related to the thermal gradient, rather than the difference of temperature between heat source and sink. The role of the stack is simply to bring down temperatures in a short distance, in order to increase the thermal gradient. It is made up of low thermal-conducting porous materials, and it is placed within the heat source and cold sink. Metal foams or ceramic materials are usually adopted for this purpose. As the thermal gradient overcomes a critical value, the gas in contact with the stack becomes the seat of a vibration, which propagates all along the resonant circuit [57]. The resonant circuit can be connected to an electro-mechanical converter, yielding electric power [58,59].

The hot temperature T_h plays an important role in determining the performance of the thermoacoustic engine, as well as the quality of the available heat source, as it affects Carnot efficiency. Therefore, it is more rational to assess the performance of the system by relative Carnot efficiency (defined as the ratio of the thermal efficiency to the Carnot efficiency Equation (14)) [56].

$$\varepsilon = \left(1 - \frac{T_c}{T_h}\right) \times 100\% \quad (14)$$

3.7. Thermoacoustic—Magnetohydrodynamic Generator

In [60], a review on the state-of-the-art conversion of thermoacoustic power into electricity is given. The primary heat produced through combustion can be converted into mechanical energy by a thermoacoustic resonator (TAR), followed by the conversion of mechanical energy into electricity via various methods. Piezoelectricity, for example, is one of the possibilities, but this effortless technique produces only a little power (a maximum

of a few watts) [29,31,61]. Other solutions producing more power are represented by rotating machines, which do not meet the objective of the study, i.e., a plant without any moving part. Another possibility is using linear induction machines, which better exploit the alternate nature of the input power. Although this concept is much humbler than the previous one, this option also involves mechanical parts in motion [62].

The prospect of linking the thermoacoustic effect to the magnetohydrodynamic effect is very appealing, as it does not include any mobile mechanical parts. Some solutions, which foresee the combination of a thermoacoustic resonator with an MHD generator, have been presented in the literature [58,61–63]. In this type of solution, the impedance of the MHD generator must be adapted to the thermoacoustic resonator, especially in the case of a two-phase system, which is the only one for which experimental data are available [59]. Frequency is a fundamental parameter, because it determines the power density of the device, which is a critical factor in space and automotive applications, much less in industrial applications. It is worth noting that, regardless of the operating frequency, at equal power, the TAR + MHD system has much smaller dimensions than alternating or rotating conversion systems, because, in the former, volume forces are involved, unlike the latter in which a surface fluid–machine interaction takes place. Another aspect to consider is that the operating fluid of the MHD generator places constraints on the frequency, due to inertia. In particular, the experimentation reported in [59], in which Sodium was used as the operating fluid, suggests that the best frequency is around 12 Hz. The frequency would be even lower if Sodium was replaced with GaInSn, which has a much higher density. It is worth noting that in [63] the working fluid is an ionized gas, so the frequency limits are the same as those on the thermoacoustic resonator side. Nevertheless, no experimental results are available on this type of solution.

4. Concept Design of the Power Plant

Figure 2 shows the process concept of the proposed ZE-IGSC power plant. As a concept design, the cycle here is assumed to be ideal. This allows us to assume, in the absence of experimental data, superior performance to that of a real plant. This concerns both the efficiency of individual processes, as well as leakage and consumption for ancillary services, which are considered here as a lump sum.

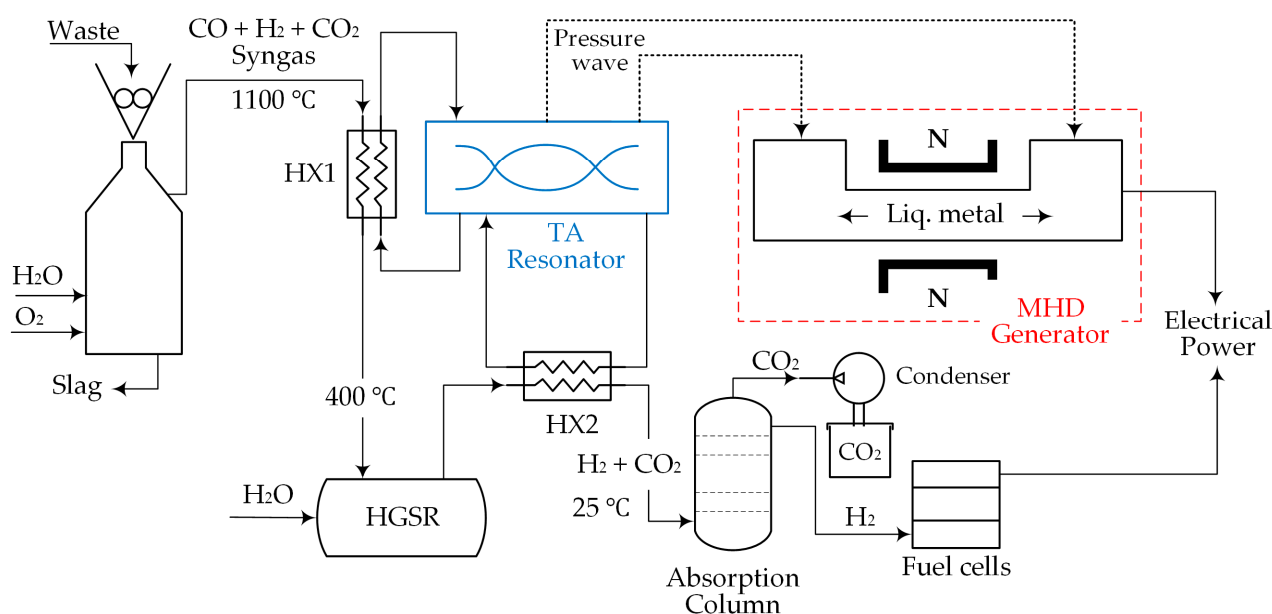


Figure 2. Concept schematic of the power plant.

Waste is broken down and then fed to the gasifier, where it reacts to produce the syngas at 1100 °C. To withstand these temperatures, the walls must be coated with materials such as tungsten carbide. The syngas enters a heat exchanger, where it provides heat to the hot side of the thermoacoustic resonator stack. In the schematics of Figure 2, an intermediate circuit—for example, a heat pipe—is foreseen, which transfers heat from the syngas to the TAR. The vibration power supplies a liquid-metal MHD generator, providing the first part of electric power. The cooled syngas (400 °C) is fed to the Water-Gas Shift Reactor (WGSR), where the mixture of CO and H₂O is converted into H₂ and CO₂. The shifted syngas enters an absorption column, where the CO₂ is separated from H₂ and stored in liquid form at room temperature (80 bar). The power consumption used to compress the CO₂ can be limited by running both the WGSR and absorption at high pressure, as the pressure does not considerably affect the shift reaction, while absorption is facilitated by high pressures. The residual syngas is almost pure H₂, and it is used to supply the Fuel Cells section, where the main quote of electrical energy of the power plant is generated.

The main components of the power plant are discussed below.

4.1. Gasifier

The gasifier in the present layout is a Waste-to-Energy (WtE) system, which uses mixed municipal solid waste (MSW) to produce heat and syngas. The MSW is crushed and fed in the WtE, and yields a hot syngas, consisting of a mixture of CO, H₂, and CO₂. A residual slag is also produced, which does not play any role in the production of electricity. It therefore increases the economic efficiency of the system, along with reducing the carbon footprint; the slag can be used as a partial replacement of cement in mortar and concrete [64,65]. If the slag is left untreated, it will have a far more adverse effect than the original MSW [66,67].

4.2. Thermoacoustic Resonator

The task of the Thermoacoustic Resonator is to exploit the enthalpy of the syngas at the outlet of the gasifier. In coal-fired IGCC plants, the temperature of the syngas is brought down by a quench, which introduces a high irreversibility into the process [23]. This quench is mandatory, because the only device devoted to convert thermal power into mechanical power is a gas turbine, and pollutants and particulate in the syngas would damage the blades. Instead, in the proposed ZE-IGSC plant, the high temperature syngas is exploited to maintain a constant temperature on the hot side of the resonator, so that the constraints on the syngas are much less strict, further contributing to improving the global efficiency of the power plant. The cold sink is represented here by a heat exchanger, where part of the waste heat of the resonator is used to supply the steam for the WGSR. Even if this solution limits the efficiency of the TAR, it represents an advantage for the global efficiency of the power plant. In the layout of the TAR, a traveling wave is adopted, which guarantees greater efficiency with respect to the alternative standing wave [6].

4.3. Magnetohydrodynamic Generator

A magnetohydrodynamic (MHD) generator is used to convert vibration power into electric power [58,61–63]. As the input power is alternated, the liquid metal type can be adopted, which allows the system to be greatly simplified. As for the liquid metal, several options are available. Often, Sodium (Na) is used for this kind of generator, due to its fluid dynamic properties, which are similar to water. However, it is solid at room temperature, and therefore it needs to be warmed up to around 400 °C. Even if the power consumption for the warming process is virtually null, as it is required only in the start-up phase, thermal isolation is required, which complicates the system. As an alternative, a Sodium-Potassium (NaK) alloy can be used, which is liquid at room temperature. However,

both solutions create problems relating to corrosion of metals and safety, due to the high reactivity of Sodium with water. Another option is represented by Gallium-Indium-Tin alloy (GalInStan), which is less reactive, but its density is much higher than that of Na, meaning its inertia is higher, which reduces the range of suitable operative frequencies. In its turn, frequency is a critical design parameter, because the higher the frequency, the higher the power density, which affects the cost of the plant. However, reducing the size of the plant, for the same power, produces several advantages, such as those related to land consumption. In [61], a conceptual design is presented, where the liquid metal is substituted with an ionized gas where the positive charges are separated from the negative ones by an external DC electrical field. Two clouds of charges are moved by a thermoacoustic wave, giving rise to two alternated currents, which perform as the primary of a transformer, where the secondary is a toroidal coil connected to the load. Resorting to a gas-phase working fluid allows the range of suitable frequencies to be extended, ultimately influencing the specific costs of the power plant.

5. Case Study

A case study in the southern part of Lebanon has been used as reference. More specifically, forty-three towns in the three Union of Municipalities of Iqleem at-Tuffah, Jezzine, and Jabal Al Rihan, encompassing area of 139 km², have been considered. The motivation of such a choice pertains to the numerous open landfill sites distributed throughout the targeted area, which exert immediate adverse effects on the well-being of residents and the surrounding ecosystem. The persistent incineration of waste materials at these open landfill sites is also posited as a significant etiological factor for various diseases and disorders. Importantly, a subset of these landfills is situated in direct proximity to the banks of the Sainiq River. Consequently, the leachates emanating from these landfills directly contaminate the underlying aquifers, the Sainiq River itself, and ultimately find their way into the Mediterranean Sea. Addressing this central environmental concern stands as the primary objective of the undertaken power plant layout [68].

The net production of mixed solid waste generated at the target area is 65 tons/day according to a study conducted in 2022 [68]. Therefore, 65 tons/day of mixed solid waste can generate energy content as follows [Equation (15)]:

$$\begin{aligned}\text{Energy content of 65 t/day} &= \text{Energy content} \times \text{Waste production} \\ &= 19.55 \text{ MJ/kg} \times 65 \text{ t/day} = 14.71 \text{ MW}\end{aligned}\quad (15)$$

Table 2 reports the overall mass and energy balance of the ZE-IGSC power plant. The layout of the power plant is shown in Figure 2. The waste is broken down and then gasified, yielding a syngas (Equations (1) and (2)) with temperature of about 1100 °C (1373 K). The gasification is assumed to be supplied with slurry, which is pre-heated using the exhaust of syngas downstream of the production of steam for the WGSR [28]. The choice of slurry has been preferred over the dry solution, because it simplifies the pressurization of the charge, and it produces the water required for the reaction. In fact, in the dry solution, a certain amount of steam must be provided to the gasifier.

The syngas exchanges heat (HX1) with the thermoacoustic resonator (TAR). The temperature of the syngas decreases progressively to the temperature of the WGSR, where it is used to produce steam for the gasifier and the shift reaction. This implies that the thermic-to-acoustic efficiency is not constant, but it decreases from a maximum of 31% at the heat exchanger inlet to a minimum of 22% at the outlet of HX1. The global efficiency of the TA-MHD generator is evaluated by integrating the value of efficiency with respect to the temperature, assuming the ambient temperature is equal to 25 °C. The electrical

power generated in this section is about 500 kW, which represents 4.6% of the total power generated in the plant.

Table 2. Mass and energy balance and the main features of the ZE-IGSC plant.

Quantity	Value
Auxiliaries power (MW)	0.58
Carbon dioxide mass flow (kg/s)	1.95
Efficiency acoustic to electric conversion (%)	80.00
Efficiency of Carnot for Thermoacoustic resonator (%)	64.57
Efficiency of thermal to acoustic conversion (%)	32.28
Efficiency of thermal to electric conversion (%)	25.83
Gasifier cold gas efficiency (%)	82.10
Heat capacity of syngas (MJ/kg K)	0.00
Liquid Metal	NaK
Mass Flow of Hydrogen (kg/s)	0.12
Mass flow of Oxygen (kg/s)	0.63
Mass flow of raw gas (kg/s)	1.46
Mass flow of steam (kg/s)	0.06
Mass flow of steam for Shift Reactor (kg/s)	1.21
Mass flow of Waste (kg/s)	0.75
Percentage of Carbon dioxide removed (%)	97.40
Percentage of CO converted (%)	97.50
Power entering into the plant (MW)	14.71
Power entering into the TA-MHD generator (MW)	1.97
Power for CO ₂ removal and compression (MW)	0.38
Power for Oxygen production and compression (MW)	2.29
Power of Fuel Cells (MW)	10.45
Power of MHD Generator (MW)	0.51
Mass flow of steam for gas treatment(kg/s)	2.54
Temperature at the outlet of Gasifier (°C)	1100
Temperature at the outlet of HX1 (°C)	400
Temperature of ambient (°C)	27.00
Net Power (MW)	7.71
Net Efficiency (%)	52.39

The syngas passing through HX1 reaches a temperature of about 400 °C (673 K), and it passes through water–gas shift reactor (WGSR) where the CO fraction is oxidized by its reaction with steam, transferring its chemical energy to the H₂. The assumed ratio of conversion is reported in Table 2. Another heat exchanger (HX2) has the task of catching any further energy lost by the WGSR. The mixture of H₂ and CO₂ is passed through an absorption column to separate CO₂ from H₂. Such a column can work through chemical or physical processes [28]. Chemical absorption is more efficient than physical absorption, but the latter requires less energy, hence its selection in the present study.

The purified H₂ is fed to fuel cells, which represent the main part of the power section (Figure 3). Fuel cell efficiency has been attributed based on the products available on the market. As an alternative to the fuel cell, other solutions were available, such as, for example, burning H₂ and using a second TA-MHD generator, but the efficiency would be much lower.

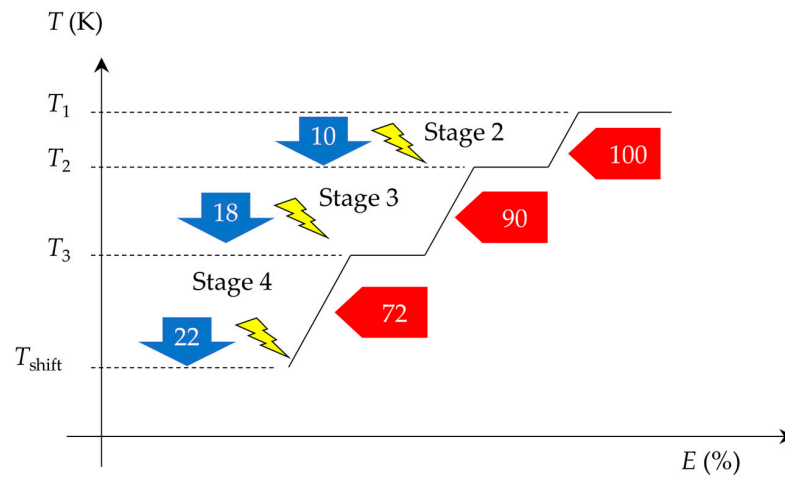


Figure 3. The energy production at each stage of fuel cell.

The separated CO₂ is pressurized at 80 bar and 25 °C (298 K). In such conditions, CO₂ is in liquid form and represents a marketable by-product.

The electric power generated is considered surplus, as the main objective is to reduce mixed solid waste in a manner that it does not release any emissions into the atmosphere.

Mass and Energy Balance

Table 3 represents the energy content carried by each type of waste in mixed municipal solid waste.

Table 3. Content of energy by waste type.

Component	% Waste by Mass	% Moisture Content	Energy Content (MJ/kg)	Dry Mass (kg)	Total Energy (MJ)
Tin cans	2	4	0.70	1.92	1.40
Food waste	15	70	3.65	4.50	54.75
Garden waste	10	50	6.60	5.00	66.00
Wood	6	20	18.60	4.80	111.60
Cardboard	10	5	14.30	9.50	143.00
Plastic	10	15	30.60	8.50	306.00
Paper	45	6	18.75	42.30	843.75
	98			76.52	1526.50

The net energy produced by 100 kg of mixed municipal waste is 1526.50 MJ (15.265 MJ/kg). To evaluate the net energy of dry mass, the percentage of moisture content (%MC) is first calculated as follows (16):

$$\begin{aligned}
 \% \text{ Moisture Content (MC)} &= \frac{\text{Total mass} - \text{Dry mass}}{\text{Total mass}} \times 100\% \\
 &= \frac{98.00 - 76.52}{98.00} \times 100\% = 21.92\%
 \end{aligned} \tag{16}$$

Then, the energy content pertaining to dry mass is evaluated as follows:

$$\begin{aligned}
 \text{Energy content} &= \text{Total energy} \times \frac{1}{1 - \%MC} \\
 &= 15.265 \text{ MJ/kg} \times \frac{1}{1 - 0.2192} = 19.550 \text{ MJ/kg}
 \end{aligned} \tag{17}$$

6. Economic Analysis

Economic analysis is complicated by the fact that there is no standard procedure for assessing the financial and external costs of waste treatment in general and incineration plants in particular. This is not helped by the literature on power generation plants using other types of fuels, because, in the present case, the ultimate goal is not to produce energy, but to solve the problem of waste management. For this reason, comparing the amount of energy obtainable from waste and other primary sources is not useful, because even if it turns out that other sources are economically more advantageous, we will still have to decide how to deal with waste. In other words, waste exerts a pressure that other primary sources lack, because while the latter can stay where they are if they are not used, waste cannot. A detailed economic analysis is beyond the scope of this paper, so the analysis was limited to the case study described in the previous section. For the methodology, the method presented in [69] has been adopted.

In [70,71] the results of several studies of WtE plants in Europe and America are reported, with sizes ranging from 40 kt/yr up to 450 kt/yr, corresponding to a specific investment cost ranging from a minimum of 430 €/ (kt/yr) to a maximum of 1030 €/ (kt/yr) and specific operating cost ranging from a minimum of 28 €/t to 48.8 €/t. It should be noted that the specific cost depends not only on the size of the plant but also on the year of construction, in the sense that more recent plants have been able to take advantage of technological innovation, which has ultimately allowed them to reduce costs. In the case under consideration in the present work, even assuming that the entire waste production in the area is allocated to thermal treatment, a plant of 23 kt/yr would be expected, which is about half of the smaller size considered in the study cited (40 kt/yr). It can be reasonably assumed that the expected cost of such a plant, i.e., CAPEX 15 M€ and OPEX 1.12 M€/yr, represents a valid reference for a plant using conventional technologies and capable of treating the waste produced in the study area.

The aim of the present study is to carry out economic analysis under the assumption that the process described in [68] is fully implemented. This envisages that only the non-recyclable fraction of the waste would be sent for thermal treatment, so the plant would be called upon, when fully operational, to treat a much smaller quantity of waste than indicated above. Nevertheless, it is assumed that the initial investment cannot be reduced by much, because there will inevitably be a minimum expenditure below which the plant cannot be built with conventional technologies. In [69], the net cost is taken into account, by considering the gains from the sale of electricity and heat. Assuming a market price of electricity of 50 €/kWh, and considering the highest efficiency recorded (15.3%), for a conventional WtE plant the revenue from the sale of electricity would amount to 0.99 M€/yr, which only partially covers operating costs. Nevertheless, a WtE plant is not required to be economically profitable, but to keep costs as low as possible, without prejudice to environmental and human protection. In other words, “do nothing” is not an option.

The technology presented in this paper has the advantage of being easily scalable, so whatever degree of waste sorting can be achieved, this will not lead to a reduction in yield. When conducting economic analysis, especially when dealing with waste, it is important to refer to both Life Cycle Assessment (LCA) and Life Cycle Costing (LCC). In the former, the air pollution, by-products disposal, and disamenities caused by the plant, directly or indirectly, over its lifetime, play a major role. It is therefore necessary to consider the so-called external costs, for which there is, admittedly, no standard methodology for calculating estimations. An interesting proposal can be found in [71], although the subject of the study is energy production plants, not waste treatment plants. In [70] the external costs of WtE plants are quantified in a very wide range (8–39 €/t) with a median of around 10 €/t. A large part is attributed to atmospheric emissions, while other aspects, such as land

consumption and disamenities, affect them to a lesser extent and, in any case, compared to the landfill alternative, incineration represents a solution with lower external costs. With regard to atmospheric emissions, it is argued in [69] that external costs, if the plant were to be built in industrial or urban areas, would be negligible as they are marginally irrelevant. We disagree with this position; if the new plant contributes to the deterioration of the already compromised environment, by the time it is decided to remedy the situation, the plant would be one more problem to solve. As far as LCC is concerned, several examples are given in [69], where incineration is preferable to separate collection. In reality, this cost item is highly dependent on how waste is collected and sorted. In fact, the economic viability of separate collection strongly depends on the commercial value of the sorted materials and how efficiently they are separated. In [68], the method adopted ensures a sorting efficiency very close to 100%, at the same time guaranteeing a high commercial value for the separated materials.

This method is based on the socialization of the separation process. The compostable fraction of the waste that has not previously been contaminated by other materials is taken away at source and is delivered for home composting or, optionally, to communal or neighbourhood composting units. This ensures that the quality of the compost is very high—specifically, that it is not contaminated with heavy metals—so that even without providing for its commercialization, its use is possible, permitting the economic advantage of not having to dispose of it. The other fractions are delivered daily to collection points located at shopping centres, where an experienced operator provides detailed separation and temporary storage. The cost of collection is reduced by the fact that the number of collection points is very limited and by the fact that the organic fraction is not collected. Finally, the high degree of separation of the individual materials means that they take on a high commercial value, contributing to cover the operating costs of collection. The model described would be economically less burdensome than both an undifferentiated collection system, because the quantities would be reduced, or a system with a door-to-door collection system, where the degree of separation detail would, in any case, be lower and the collection operations much more burdensome.

Ultimately, the technological solution adopted plays a key role in establishing its cost-effectiveness, it being understood that the sustainability of a solution must be ensured in both an economic and environmental sense, i.e., LCA and LCC analyses must be considered simultaneously. The objective of this study is to reduce the standard cost of a WtE plant and at the same time eliminate emissions into the atmosphere. The use of gasification for waste treatment has been proposed in [72], demonstrating the efficiency and cost-effectiveness of this solution. The investment cost is lower than that of the incinerator (12 M€), while the operating costs are slightly higher (1.38 M€/yr). The return in this case is higher than that of the incinerator (27%), so there is a positive balance between operating costs and revenues, but these are still not sufficient to cover the capital depreciation, assuming a 5% discount rate.

In the case of the proposed plant, due to its high efficiency (52.39%), the annual revenues for commercialization of electrical energy are sufficient to cover the operating costs and the increase for the discount rate, leading to a payback time of just over six years. A few considerations warrant mentioning. The complete elimination of atmospheric emissions reduces external costs, although the extent of this reduction is particularly complex to assess and is beyond the scope of this work. As far as investment costs are concerned, compared to other waste gasification plants, there are two extra components in the proposed layout, i.e., the WGS and absorption column, but they allow the downstream mass flow to be greatly reduced, as it is limited to hydrogen only, and this results in a reduction in the piping cost.

Lastly, there is a lack of literature on the costs of the conversion system based on thermoacoustic-magnetohydrodynamic coupling, particularly with regard to the induction-type generator. It is, however, possible to make a few considerations. Most electrical power is generated by the fuel cell, while the TA-MHD generator produces just 5% of the total power. The decision to include this power section was aimed at limiting the irreversibility of syngas quench, which is the standard in IGCC plants [38]. However, should the cost of the device prove to be unjustified, the effect on the plant's overall performance would, in any case, be minimal. Nonetheless, the absence of solid moving parts and of electrodes for drawing current, the characteristic down-scalability of this type of system, as well as its capability to deal with dirty gas and very high temperatures, lead one to assume that there would also be an economic case for its inclusion, hence the choice made.

Interesting insights into the economics, albeit referring to a different type of plant, can be drawn from the study conducted in [73], where the liquid metal MHD generator is used in a concentrating solar power (CSP) plant to produce electricity, hydrogen, and fresh water.

7. Discussion and Conclusions

The integration of thermoacoustic resonators (TAR) and magnetohydrodynamic (MHD) systems into a waste-to-energy (WtE) power plant represents a transformative approach to sustainable energy generation, offering a high-efficiency, low-emission alternative to conventional waste incineration and fossil fuel-based power generation. This innovative system capitalizes on the synergistic potential of thermochemical, acoustic, and electrodynamic energy conversion processes to maximize energy extraction from waste, while minimizing environmental impact. By converting municipal and industrial waste into syngas, harnessing its thermal energy through thermoacoustic pressure waves, and subsequently driving an MHD generator with liquid NaK, the system achieves a dual-stage electricity generation process that significantly outperforms traditional Rankine or Brayton cycles in terms of efficiency and scalability. Furthermore, the integration of a water-gas shift reactor (WGSR) and hydrogen fuel cells adds a secondary layer of energy recovery, while the capture and liquefaction of CO₂ align with global decarbonization objectives. The implications of this technology extend beyond waste management, offering a blueprint for next-generation renewable energy systems that prioritize circular economy principles and carbon neutrality.

One of the most compelling advantages of this system lies in its thermodynamic efficiency. Traditional WtE plants typically suffer from significant energy losses due to mechanical inefficiencies in turbines and heat exchangers. In contrast, the TAR-MHD configuration eliminates moving parts in the primary energy conversion stage, relying instead on the direct interaction between acoustic waves and the conductive fluid within the MHD generator. This not only reduces mechanical wear and maintenance costs but also enhances the system's adaptability to fluctuating thermal inputs, a common challenge in waste-derived syngas combustion. Experimental studies on standalone TAR and MHD systems have demonstrated their potential for achieving conversion efficiencies exceeding 40% under optimal conditions, suggesting that their integration could yield even higher performance when optimized for large-scale applications.

From an environmental perspective, the proposed system addresses two critical challenges: waste accumulation and greenhouse gas emissions. By gasifying waste rather than incinerating it outright, the process reduces the release of harmful pollutants such as dioxins and particulate matter, while the WGSR ensures that the resulting syngas is enriched with hydrogen—a clean energy carrier. The subsequent condensation of CO₂ into a liquid form not only mitigates emissions but also creates opportunities for carbon

utilization in industrial processes or long-term sequestration. This dual focus on energy recovery and emission reduction positions the technology as a viable candidate for meeting stringent climate targets, particularly in urban areas where waste management and clean energy generation are pressing concerns.

However, the practical implementation of this system is not without its challenges. The high operating temperatures required for efficient thermoacoustic conversion and MHD operation demand advanced materials capable of withstanding thermal and chemical degradation. Corrosion-resistant alloys and ceramic composites will be essential for the reactor walls, heat exchangers, and MHD channels, particularly when handling aggressive media like molten NaK and syngas. Additionally, the system's economic viability hinges on scaling up laboratory-scale prototypes to industrial levels, where factors such as capital costs, energy output, and grid compatibility must be carefully evaluated. Pilot projects and techno-economic analyses will be critical in determining whether the TAR-MHD WtE plant can compete with established renewable energy technologies such as wind, solar, and conventional WtE incineration.

Beyond waste-to-energy applications, the principles underlying this hybrid system have far-reaching implications for renewable energy innovation. The ability of TARs to convert low-grade heat into usable acoustic energy makes them suitable for integration with other thermal sources, such as concentrated solar power (CSP) or industrial waste heat recovery systems. Similarly, MHD generators could be adapted for use in geothermal or nuclear power plants, where their lack of moving parts would offer reliability advantages over traditional turbines. The modularity of both technologies further enhances their appeal, enabling deployment in decentralized energy systems where grid connectivity is limited.

Looking ahead, future research should prioritize several key areas to unlock the full potential of this technology. First, experimental validation of integrated TAR-MHD systems at pilot scales is essential to identify and address operational challenges, such as acoustic damping and magnetic field uniformity in the MHD generator. Second, advancements in materials science, particularly in the development of high-temperature superconductors and corrosion-resistant coatings, could dramatically improve system longevity and efficiency. Third, the integration of artificial intelligence for real-time optimization of thermoacoustic and MHD parameters could enhance performance under variable load conditions. Finally, interdisciplinary collaboration between acousticians, plasma physicists, and chemical engineers will be crucial in refining the system's design and ensuring its compatibility with existing energy infrastructure.

In summary, the proposed TAR-MHD WtE power plant exemplifies the convergence of sustainability and innovation, offering a scalable and efficient solution to the dual crises of waste management and energy scarcity. By leveraging the unique properties of thermoacoustics and magnetohydrodynamics, the system not only maximizes energy recovery from waste but also sets a precedent for future renewable energy technologies that prioritize efficiency, environmental stewardship, and circular economy principles. While technical and economic hurdles remain, the system's potential to revolutionize both the waste and energy sectors underscores the importance of continued investment and research in this emerging field. As global efforts to combat climate change intensify, technologies like this will play an increasingly vital role in the transition to a carbon-neutral future.

Author Contributions: Conceptualization, A.M. and A.R.; methodology, A.M. and A.R.; software, A.R.; validation, A.R.; formal analysis, A.M. and A.R.; investigation, A.M., and A.R.; resources, A.M. and A.R.; data curation, A.R.; writing—original draft preparation, A.R.; writing—review and editing, A.M. and A.R.; visualization, A.R.; supervision, A.M.; project administration, A.M.; funding acquisition, A.M. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: No new data were created.

Conflicts of Interest: The authors declare no conflicts of interest.

List of Nomenclature

Quantity (Symbol)	SI Unit
Bulk viscosity (ζ)	Pa·s
Charge (q)	C
Current density (J)	A/m ²
Density (ρ)	kg/m ³
Efficiency (ε)	
Electric conductivity (σ)	S/m
Electrical power (P_{el})	W
Fluid dynamic viscosity (η)	Pa·s
Generic internal power source (S_{int})	
Gravitational acceleration (g)	m/s ²
Heat capacity ratio (γ)	
Identity tensor (I_d)	
Load factor ratio (K)	
Lorentz force (F)	N
Magnetic field (B)	T
Open circuit voltage (V_0)	V
Pressure (p)	Pa
Specific gas constant (R)	J/(K·mol)
Specific gravity force (ρg)	N/m ³
Specific heat at constant pressure (c_p)	J/(kg·K)
Specific heat at constant volume (c_v)	J/(kg·K)
Standard molar formation enthalpy of X ($\Delta H_{fX}^\ominus$)	kJ/mol
Stress tensor (τ)	Pa
Temperature (T)	K
Temperature of cold source (T_c)	K
Temperature of hot source (T_h)	K
Thermal conductivity (k)	W/(m·K)
Total energy per volume unit (e)	J/m ³
Velocity (v)	m/s
Wall heat transfer per volume unit and time unit (Q_{wall})	W/m ³

References

1. Swilling, M.; Hajer, M.; Baynes, T.; Bergesen, J.; Labbé, F.; Musango, J.K.; Ramaswami, A.; Robinson, B.; Salat, S.; Suh, S.; et al. *The Weight of Cities: Resource Requirements of Future Urbanization*; International Resource Panel: Paris, France; United Nations Environment Programme: Nairobi, Kenya, 2014; p. 280.
2. Peter, C.; Swilling, M. *Sustainable, Resource Efficient Cities-Making It Happen!* United Nations Environment Programme: Paris, France, 2012; p. 64.
3. Bouton, S.; Cis, D.; Mendonca, L.; Pohl, H.; Remes, J.; Ritchie, H.; Woetzel, J. *How to Make a City Great*; McKinsey Cities Special Initiative, McKinsey & Company: Chicago, IL, USA, 2013; p. 35.
4. Liu, C.; Kawamoto, K.; Sasaki, S. *Waste-to-Energy Incineration*; CCET guideline series on intermediate municipal solid waste treatment technologies; United Nations Environment Programme: Nairobi, Kenya; International Environmental Technology Centre: Osaka, Japan; Japan Society of Material Cycles and Waste Management: Tokyo, Japan; Institute for Global Environmental Strategies: Hayama, Japan, 2020; p. 45.

5. Makarichi, L.; Jutidamrongphan, W.; anan Techato, K. The Evolution of Waste-to-Energy Incineration: A Review. *Renew. Sustain. Energy Rev.* **2018**, *91*, 812–821. [CrossRef]
6. Astrup, T.F.; Tonini, D.; Turconi, R.; Boldrin, A. Life Cycle Assessment of Thermal Waste-to-Energy Technologies: Review and Recommendations. *Waste Manag.* **2015**, *37*, 104–115. [CrossRef] [PubMed]
7. Baeyens, J.; Van Puyvelde, F. Fluidized Bed Incineration of Sewage Sludge: A Strategy for the Design of the Incinerator and the Future for Incinerator Ash Utilization. *J. Hazard. Mater.* **1994**, *37*, 179–190. [CrossRef]
8. Tan, S.T.; Lee, C.T.; Hashim, H.; Ho, W.S.; Lim, J.S. Optimal Process Network for Municipal Solid Waste Management in Iskandar Malaysia. *J. Clean. Prod.* **2014**, *71*, 48–58. [CrossRef]
9. Khan, A.H.; López-Maldonado, E.A.; Alam, S.S.; Khan, N.A.; López, J.R.L.; Herrera, P.F.M.; Abutaleb, A.; Ahmed, S.; Singh, L. Municipal Solid Waste Generation and the Current State of Waste-to-Energy Potential: State of Art Review. *Energy Convers. Manag.* **2022**, *267*, 115905. [CrossRef]
10. Zaman, A.U. A Comprehensive Review of the Development of Zero Waste Management: Lessons Learned and Guidelines. *J. Clean. Prod.* **2015**, *91*, 12–25. [CrossRef]
11. Nagar, V.; Kaushal, R. A Review of Recent Advancement in Plasma Gasification: A Promising Solution for Waste Management and Energy Production. *Int. J. Hydrogen Energy* **2024**, *77*, 405–419. [CrossRef]
12. Hinkel, M.; Blume, S.; Hinchliffe, D.; Mutz, D.; Hengevoss, D. *Guidelines on Pre-and Co-Processing of Waste in Cement Production: Use of Waste as Alternative Fuel and Raw Material*; Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ) GmbH LafargeHolcim Institut für Ecopreneurship, Hochschule für Life Sciences FHNW: Holderbank, Switzerland, 2020; p. 132.
13. National Research Council (US); Committee on Health Effects of Waste Incineration Processes and Environmental Releases. *Waste Incineration and Public Health*; National Academies Press: Washington, DC, USA, 2000; pp. 34–70; ISBN 978-0-309-06371-5.
14. Tempelman, E.; Shercliff, H.; van Eyben, B.N. Recycling. In *Manufacturing and Design*; Elsevier: Amsterdam, The Netherlands; Butterworth-Heinemann: Oxford, UK, 2014; pp. 251–267; ISBN 978-0-08-099922-7.
15. Zhu, Y.; Frey, H.C. Integrated Gasification Combined Cycle (IGCC) Systems. In *Combined Cycle Systems for Near-Zero Emission Power Generation*; Rao, A.D., Ed.; Woodhead Publishing Limited: Soston, UK, 2012; pp. 129–161; ISBN 978-0-85709-013-3.
16. McCoy, J.T.; Auret, L. Machine Learning Applications in Minerals Processing: A Review. *Miner. Eng.* **2019**, *132*, 95–109. [CrossRef]
17. Gomez-Flores, A.; Ilyas, S.; Heyes, G.W.; Kim, H. A Critical Review of Artificial Intelligence in Mineral Concentration. *Miner. Eng.* **2022**, *189*, 107884. [CrossRef]
18. National Fire Protection Association (NFPA). *NFPA 120: Standard for Fire Prevention and Control in Coal Mines*; National Fire Protection Association (NFPA): Quincy, MA, USA, 2023.
19. Jiang, B.; Liu, Z.; Zhao, Y.; Zhang, X.; Wang, X.-H.; Ji, B.; Zhang, Y.; Huang, J. Development of an Eco-Friendly Dust Suppressant Based on Modified Pectin: Experimental and Theoretical Investigations. *Energy* **2024**, *289*, 130018. [CrossRef]
20. Salam, A. Internet of Things for Sustainable Mining. In *Internet of Things for Sustainable Community Development: Wireless Communications, Sensing, and Systems*; Salam, A., Ed.; Springer International Publishing: Cham, Switzerland, 2020; pp. 243–271; ISBN 978-3-030-35291-2.
21. Ramesh, V. IoT in Mining Solutions and Its Innovations. Available online: <https://www.infosysbpm.com/blogs/sourcing-procurement/iot-in-mining.html> (accessed on 26 March 2025).
22. Lodewijks, G.; Li, W.; Pang, Y.; Jiang, X. An Application of the IoT in Belt Conveyor Systems. In *Proceedings of the Internet and Distributed Computing Systems*; Li, W., Ali, S., Lodewijks, G., Fortino, G., Di Fatta, G., Yin, Z., Pathan, M., Guerrieri, A., Wang, Q., Eds.; Springer International Publishing: Cham, Switzerland, 2016; pp. 340–351.
23. Williams, G. Social and Economic Costs and Benefits of Coal. In *The Coal Handbook (Second Edition)*; Osborne, D., Ed.; Woodhead Publishing Series in Energy; Woodhead Publishing: Soston, UK, 2023; Volume 2, pp. 3–29; ISBN 978-0-12-824327-5.
24. Zagorščak, R.; Metcalfe, R.; Limer, L.; Thomas, H.; An, N.; Bond, A.; Watson, S. Risk Assessment Methodology for Underground Coal Gasification Technology. *J. Clean. Prod.* **2022**, *370*, 133493. [CrossRef]
25. Swift, G.W. *Thermoacoustics: A Unifying Perspective for Some Engines and Refrigerators*, 2nd ed.; Springer Nature: Cham, Switzerland, 2017; ISBN 978-3-319-66933-5.
26. Falco, M.D.; Iaquaniello, G.; Centi, G. (Eds.) CO₂: A Valuable Source of Carbon. In *Green Energy and Technology*, 1st ed.; Springer: London, UK, 2013; ISBN 978-1-4471-5119-7.
27. Song, C. CO₂ Conversion and Utilization: An Overview. In *ACS Symposium Series*; American Chemical Society: Washington, DC, USA, 2002; Volume 809, pp. 1–30; ISBN 978-0-8412-1907-6.
28. Cau, G.; Cocco, D.; Montisci, A. Performance of Zero Emissions Integrated Gasification Hydrogen Combustion (ZE-IGHC) Power Plants with CO₂ Removal. In *Proceedings of the ASME Turbo Expo 2001: Power for Land, Sea and Air*, New Orleans, LA, USA, 4–7 June 2001; American Society of Mechanical Engineers (ASME): New Orleans, LA, USA, 2001; Volume 2, p. 12.
29. Bell, D.A.; Towler, B.F.; Fan, M. *Coal Gasification and Its Applications*, 1st ed.; William Andrew: Norwich, NY, USA, 2010. ISBN 978-0-8155-2049-8.

30. National Institute of Standards and Technology. Carbon Monoxide. Available online: <https://webbook.nist.gov/cgi/cbook.cgi?ID=C630080&Units=SI&Mask=1#Thermo-Gas> (accessed on 9 July 2021).
31. National Institute of Standards and Technology. Carbon Dioxide. Available online: <https://webbook.nist.gov/cgi/cbook.cgi?ID=C124389&Units=SI&Mask=1> (accessed on 29 June 2021).
32. National Institute of Standards and Technology. Water. Available online: <https://webbook.nist.gov/cgi/cbook.cgi?ID=C7732185&Units=SI&Mask=1#Thermo-Gas> (accessed on 29 June 2021).
33. Idriss, H.; Scott, M.; Subramani, V. Introduction to Hydrogen and Its Properties. In *Compendium of Hydrogen Energy*; Subramani, V., Basile, A., Veziroğlu, T.N., Eds.; Elsevier: Amsterdam, The Netherlands, 2015; pp. 3–19; ISBN 978-1-78242-361-4.
34. Gokhale, A.A.; Dumesic, J.A.; Mavrikakis, M. On the Mechanism of Low-Temperature Water Gas Shift Reaction on Copper. *J. Am. Chem. Soc.* **2008**, *130*, 1402–1414. [[CrossRef](#)]
35. National Energy Technology Laboratory. Water Gas Shift & Hydrogen Production. Available online: <https://www.netl.doe.gov/research/coal/energy-systems/gasification/gasifiedia/water-gas-shift> (accessed on 9 July 2021).
36. Cormos, C.C.; Cormos, A.M.; Serban Agachi, P. Techno-Economical and Environmental Evaluations of IGCC Power Generation Process with Carbon Capture and Storage (CCS). *Comput. Aided Chem. Eng.* **2011**, *29*, 1678–1682. [[CrossRef](#)]
37. Mitsubishi Power. Integrated Coal Gasification Combined Cycle (IGCC) Power Plants. Available online: <https://power.mhi.com/products/igcc> (accessed on 22 June 2021).
38. Maurstad, O. *An Overview of Coal Based Integrated Gasification Combined Cycle (IGCC) Technology*; Massachusetts Institute of Technology, Laboratory for Energy and the Environment: Cambridge, MA, USA, 2005; p. 44.
39. Capodaglio, A.G.; Bolognesi, S. Ecofuel Feedstocks and Their Prospects. In *Advances in Eco-Fuels for a Sustainable Environment*; Elsevier: Amsterdam, The Netherlands; Woodhead Publishing: Soston, UK, 2019; pp. 15–51; ISBN 978-0-08-102728-8.
40. Moya, D.; Aldás, C.; Jaramillo, D.; Játiva, E.; Kaparaju, P. Waste-To-Energy Technologies: An Opportunity of Energy Recovery from Municipal Solid Waste, Using Quito-Ecuador as Case Study. *Energy Procedia* **2017**, *134*, 327–336. [[CrossRef](#)]
41. Sheindlin, A.E.; Jackson, W.D.; Brzozowski, W.S.; Rietjens, L.H.T. Magnetohydrodynamic Power Generation. *Nat. Resour. Forum* **1979**, *3*, 133–145. [[CrossRef](#)]
42. Bera, T.K. A Magnetohydrodynamic (MHD) Power Generating System: A Technical Review. *IOP Conf Ser. Mater. Sci. Eng.* **2020**, *955*, 012075. [[CrossRef](#)]
43. Messerle, H.K. *Magnetohydrodynamic Electrical Power Generation (UNESCO Energy Engineering Series)*; John Wiley Publishing: Chichester, UK, 1995; ISBN 978-0-471-94252-8.
44. Forcinetti, R. *Magneto-Fluid-Dynamic Energy Conversion Systems for Aerospace Applications*; Università degli Studi di Cagliari: Cagliari, Italy, 2017.
45. Lani, A.; Sharma, V.; Giangaspero, V.F.; Poedts, S.; Viladegut, A.; Chazot, O.; Giacomelli, J.; Oswald, J.; Behnke, A.; Pagan, A.S.; et al. A Magnetohydrodynamic Enhanced Entry System for Space Transportation: MEESS. *J. Space Saf. Eng.* **2023**, *10*, 27–34. [[CrossRef](#)]
46. Kayukawa, N. Open-Cycle Magnetohydrodynamic Electrical Power Generation: A Review and Future Perspectives. *Prog. Energy Combust. Sci.* **2004**, *30*, 33–60. [[CrossRef](#)]
47. Carcangiu, S.; Fanni, A.; Montisci, A. Optimal Design of an Inductive MHD Electric Generator. *Sustainability* **2022**, *14*, 16457. [[CrossRef](#)]
48. Carcangiu, S.; Forcinetti, R.; Montisci, A. Simulink Model of an Inductive MHD Generator. *Magnetohydrodynamics* **2017**, *53*, 255–266. [[CrossRef](#)]
49. Carcangiu, S.; Montisci, A.; Pintus, R. Performance Analysis of an Inductive MHD Generator. *Magnetohydrodynamics* **2012**, *48*, 115–124. [[CrossRef](#)]
50. Carcangiu, S.; Montisci, A. Assessment of the Machine Parameters Affecting the Overall Performance of an Inductive MHD Generator. In Proceedings of the 2012 IEEE International Energy Conference and Exhibition (ENERGYCON), Florence, Italy, 10 November 2012; pp. 271–275.
51. Rosa, R.J.; Krueger, C.H.; Shioda, S. Plasmas in MHD Power Generation. *IEEE Trans. Plasma Sci.* **1991**, *19*, 1180–1190. [[CrossRef](#)]
52. Strohl, G.R.; Jackson, W.D. Magnetohydrodynamic Power Generator. *Encycl. Br.* 2016. Technology. Available online: <https://www.britannica.com/technology/magnetohydrodynamic-power-generator> (accessed on 26 March 2025).
53. Medin, S.A. Magnetohydrodynamic Electrical Power Generators. Available online: <https://www.thermopedia.com/content/934/> (accessed on 29 June 2021).
54. Abdoulla-Latiwish, K.O.A.; Jaworski, A.J. Two-Stage Travelling-Wave Thermoacoustic Electricity Generator for Rural Areas of Developing Countries. *Appl. Acoust.* **2019**, *151*, 87–98. [[CrossRef](#)]
55. Agarwal, H.; Unni, V.R.; Akhil, K.T.; Ravi, N.T.; Iqbal, S.M.; Sujith, R.I.; Pesala, B. Compact Standing Wave Thermoacoustic Generator for Power Conversion Applications. *Appl. Acoust.* **2016**, *110*, 110–118. [[CrossRef](#)]
56. Yang, R.; Wang, Y.; Jin, T.; Feng, Y.; Tang, K. Development of a Three-Stage Looped Thermoacoustic Electric Generator Capable of Utilizing Heat Source below 120 °C. *Energy Convers. Manag.* **2018**, *155*, 161–168. [[CrossRef](#)]
57. Bi, T.; Wu, Z.; Zhang, L.; Yu, G.; Luo, E.; Dai, W. Development of a 5 kW Traveling-Wave Thermoacoustic Electric Generator. *Appl. Energy* **2017**, *185*, 1355–1361. [[CrossRef](#)]

58. Brekis, A.; Alemany, A.; Alemany, O.; Montisci, A. Space Thermoacoustic Radioisotopic Power System, SpaceTRIPS: The Magnetohydrodynamic Generator. *Sustainability* **2021**, *13*, 13498. [\[CrossRef\]](#)
59. Hofler, T.J.; Ceperley, P.H. Thermoacoustic Heat Engines. In *CRC Handbook of Thermoelectrics*; Rowe, D.M., Ed.; CRC Press: London, UK, 2012; pp. 581–601.
60. Timmer, M.A.G.; de Blok, K.; Meer, T.H. van der Review on the Conversion of Thermoacoustic Power into Electricity. *J. Acoust. Soc. Am.* **2018**, *143*, 857. [\[CrossRef\]](#)
61. Mirhoseini, S.M.H.; Alemany, A. Analytical Study of Thermoacoustic MHD Generator. *Magnetohydrodynamics* **2015**, *51*, 519–530. [\[CrossRef\]](#)
62. Alemany, A.; Carcangiu, S.; Forcinetti, R.; Montisci, A.; Roux, J.P. Magnetohydrodynamics, Feasibility Analysis of an MHD Inductive Generator Coupled with a Thermoacoustic Resonator. *Magnetohydrodynamics* **2015**, *51*, 531–542. [\[CrossRef\]](#)
63. Alemany, A.; Krauze, A.; Al Radi, M. Thermo Acoustic-MHD Electrical Generator. *Energy Procedia* **2011**, *6*, 92–100. [\[CrossRef\]](#)
64. Czop, M.; Lazniewska-Piekarczyk, B. Use of Slag from the Combustion of Solid Municipal Waste as A Partial Replacement of Cement in Mortar and Concrete. *Materials* **2020**, *13*, 1593. [\[CrossRef\]](#) [\[PubMed\]](#)
65. Zeng, C.; Lyu, Y.; Wang, D.; Ju, Y.; Shang, X.; Li, L. Application of Fly Ash and Slag Generated by Incineration of Municipal Solid Waste in Concrete. *Adv. Mater. Sci. Eng.* **2020**, *2020*, 7802103. [\[CrossRef\]](#)
66. Wang, Y.n.; Wang, Q.; Li, Y.; Wang, H.; Gao, Y.; Sun, Y.; Wang, B.; Bian, R.; Li, W.; Zhan, M. Impact of Incineration Slag Co-Disposed with Municipal Solid Waste on Methane Production and Methanogens Ecology in Landfills. *Bioresour. Technol.* **2023**, *377*, 128978. [\[CrossRef\]](#) [\[PubMed\]](#)
67. Rollinson, A.N.; Vahk, J.; Oliveira, A. *Toxic Fallout-Waste Incinerator Bottom Ash in a Circular Economy*; Zero Waste Europe, Global Alliance for Incinerator Alternatives: Brussels, Belgium, 2022; p. 26.
68. Rashid, A.; Aleik, K.; Alloush, Z.; Awada, N.; Baraki, M.; Bardus, M.; Chaar, R.; Chehade, B.; Fakih, B.; Foddiss, L.M.; et al. RES-Q an Ongoing Project on Municipal Solid Waste Management Program for the Protection of the Saniq River Basin in Southern Lebanon. In Proceedings of the 21st International Conference Computational Science and Its Applications (ICCSA 2021), Cagliari, Italy, 13–16 September 2021; Lecture Notes in Computer Science. Gervasi, O., Murgante, B., Misra, S., Garau, C., Blečić, I., Taniar, D., Apduhan, B.O., Rocha, A.M.A.C., Tarantino, E., Torre, C.M., Eds.; Springer: Cham, Switzerland, 2021; Volume 12956, pp. 536–550.
69. Massarutto, A. Economic Aspects of Thermal Treatment of Solid Waste in a Sustainable WM System. *Waste Manag.* **2015**, *37*, 45–57. [\[CrossRef\]](#) [\[PubMed\]](#)
70. Eshet, T.; Ayalon, O.; Shechter, M. Valuation of Externalities of Selected Waste Management Alternatives: A Comparative Review and Analysis. *Resour. Conserv. Recycl.* **2006**, *46*, 335–364. [\[CrossRef\]](#)
71. University of Stuttgart ExternE-External Costs of Energy. Available online: <https://www.ier.uni-stuttgart.de/en/research/projects/externe> (accessed on 29 April 2025).
72. Murphy, J.D.; McKeogh, E. Technical, Economic and Environmental Analysis of Energy Production from Municipal Solid Waste. *Renew. Energy* **2004**, *29*, 1043–1057. [\[CrossRef\]](#)
73. Nezhad, Q.A.; Jafarmadar, S.; Genceli, H. Analysis of a Novel Concentrated Solar Power and Magnetohydrodynamic Liquid Metal Units Integrated System with Hydrogen Production. *Int. J. Hydrogen Energy* **2023**, *48*, 22734–22751. [\[CrossRef\]](#)

Disclaimer/Publisher’s Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.